

Experiment No. 2

Experiment Name

Contrast Enhancement and Histogram-Based Image Processing using Python and OpenCV

Aim

To analyze and implement contrast enhancement techniques using histogram analysis and histogram equalization in Python and OpenCV for improving image quality and enhancing visual information in low-contrast images.

Purpose

Image contrast plays a significant role in the performance of computer vision systems. Poor illumination, sensor limitations, or environmental conditions often result in low-contrast images, making feature extraction and object recognition difficult. Histogram-based enhancement techniques improve the distribution of pixel intensities, thereby enhancing image visibility and facilitating subsequent image analysis tasks.

In this experiment, students will implement contrast stretching, histogram analysis, and histogram equalization using Python and OpenCV. The effectiveness of these enhancement techniques will be evaluated on real-world images such as medical images, satellite images, or low-light photographs.

Prerequisites

- Basic knowledge of digital image processing
- Familiarity with Python programming
- Python libraries: OpenCV (cv2), NumPy, Matplotlib
- Understanding of grayscale images and pixel intensity distributions
- Jupyter Notebook, Google Colab, or Visual Studio Code

Steps

1. Import the required libraries, including OpenCV, NumPy, and Matplotlib.
2. Load a low-contrast grayscale image or convert a color image into grayscale for processing.
3. Display the original image and generate its histogram to analyze the distribution of pixel intensity values.
4. Perform contrast stretching to improve the dynamic range of pixel intensities.

5. Apply Histogram Equalization using OpenCV and compare the enhanced image with the original image.
6. Implement Contrast Limited Adaptive Histogram Equalization (CLAHE) and observe its performance on images with varying illumination.
7. Plot and compare the histograms of the original, histogram-equalized, and CLAHE-enhanced images.
8. Analyze the visual improvements in terms of brightness, contrast, and feature visibility.
9. Compare the effectiveness of Histogram Equalization and CLAHE for different types of images.
10. Document the observations and discuss the role of histogram-based enhancement in computer vision applications.

Questions

1. What is image contrast, and why is it important in image processing?
2. Explain the concept of an image histogram. What information does it provide?
3. Differentiate between Histogram Stretching and Histogram Equalization.
4. What is Contrast Limited Adaptive Histogram Equalization (CLAHE)? How does it differ from standard Histogram Equalization?
5. Why is histogram equalization commonly applied before feature extraction and image segmentation?
6. Mention three real-world applications where histogram equalization is widely used.
7. What are the limitations of global histogram equalization?
8. How does contrast enhancement improve the performance of object detection and recognition systems?
9. Compare histogram-based enhancement techniques with brightness adjustment methods.
10. Why is CLAHE preferred for medical imaging and low-light image enhancement?